Coordinate Geometry

Questions 1–75

1.03 — Question 1. Find the line passing through each of the following pairs of points giving your answer in the form $ax + by = c$ where a, b and c are integers:

a) $(2, 5)$ and $(1, 7)$
b) $(1, 4)$ and $(4, 13)$
c) $(-1, -3)$ and $(5, 1)$
d) $(-4, -6)$ and $(-2, -17)$
e) (p, q) and $(2p, 4q)$
f) $(p, \frac{1}{p})$ and $(\frac{1}{p}, p)$ where $p \neq -1, 0, 1$

1.03 — Question 2. State both the gradient of the line and the gradient of a line which is perpendicular to each of the following:

a) $3x + 6y = 7$
b) $3y - 2x = 1$
c) $2x + 8y = 11$
d) $7y - 5x - 2 = 0$
e) $20x + 17y = 3$
f) $ax + by = c$

1.03 — Question 3. For each of the following pairs of lines, determine if they are perpendicular to each other:

a) $2x + 3y = 1$ and $y = -\frac{3}{2}x + 5$
b) $x - 5y + 10 = 0$ and $y = 5x - 4$
c) $24x + 12y = 16$ and $12y + 24x = 1$
d) $y = -4x + 8$ and $4y - x = 28$
e) $ax + by = p$ and $bx - ay = q$

1.03 — Question 4. For the lines $ax + by = c$ and $px + qy = r$ where a, b, c, p, q and r are integers find a relationship between a, b, p and q such that the two lines are

a) Parallel.
b) Perpendicular.

1.03 — Question 5. Find the midpoint between each of these pairs of points:

a) $(-4, 3)$ and $(4, 9)$
b) $(1, 7)$ and $(6, -5)$
c) $(-8, 10)$ and $(-2, -1)$
d) $(-7, -2)$ and $(\frac{7}{2}, 12)$
e) (p, q) and $(2p, 3q)$
f) $(p, \frac{1}{p})$ and $(\frac{1}{p}, p)$ where $p \neq -1, 0, 1$

1.03 — Question 6. For each of the following, find the value of k given the point which lies on the line:

a) $(k, 4)$ and $2y - 5x = 16$
b) $(2k, k)$ and $4x - 3y + 20 = 0$

1.03 — Question 7. Find the perpendicular bisector of the each of the following points giving your answer in the form $ax + by = c$ where a, b and c are integers:

a) $(-2, 5)$ and $(6, 1)$
b) $(3, 5)$ and $(6, 8)$
c) $(1, 2)$ and $(4, 10)$
d) $(-6, 2)$ and $(-1, -5)$
e) $(-9, 11)$ and $(-1, -2)$
f) (p, q) and $(2p, 3q)$

1.03 — Question 8. Find the equation of the straight line which is parallel to the line $2y - x = 5$ and passes through the point $(-1, 4)$. Give your answer in the form $ax + by = c$ where a, b and c are integers.

1.03 — Question 9. Find the equation of the straight line which is perpendicular to the line $6y + 2x - 11 = 0$ and passes through the point $(2, 3)$. Give your answer in the form $ax + by = c$ where a, b and c are integers.

1.03 — Question 10. Find the angle (in degrees to 1 decimal place) between each of the following pairs of lines:

a) $y = 0$ and $y = 2x$
b) $y = 3x + 3$ and $y = 4x - 1$
c) $y = 5x + 8$ and $y = -x$
d) $y = -2x - 4$ and $y = -5x + 1$

1.03 — Question 11. Find the area of the triangle bounded by the coordinate axes and the line $2y + x - 11 = 0$.

1.03 — Question 12. Find the area in terms of k of the triangle bounded by the coordinate axes and the line $ky + 3x = 2$ where k is a positive constant.

1.03 — Question 13. The three points $(3, 11)$, $(4, -2)$ and $(-2, 1)$ are all corners of a triangle. By considering the gradients of the lines passing through each of the pairs of points, show that it is a right-angled triangle.

1.03 — Question 14. The three points $(5, 10)$, $(-1, 8)$ and $(4, -7)$ are all corners of a triangle. By considering the distances between each of the pairs of points, show that it is a right-angled triangle and hence find the area of the triangle.

1.03 — Question 15. Find the value of k such that $2y - 7x = 0$ and $ky + 6x = 10$ are perpendicular to each other.

1.03 — Question 16. A line perpendicular to $y + 2x = 9$ passes through $(p, 6)$ and $(1, 7)$. Find the value of p.

1.03 — Question 17. Two points $A(p, -2)$ and $B(5, q)$ have a perpendicular bisector with equation $3y + x = 15$. Find the values of p and q.

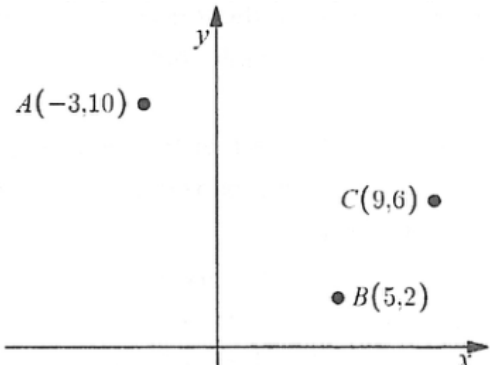
1.03 — Question 18. Show that $(-5, 7)$, $(10, 10)$ and $(-15, 5)$ are collinear.

1.03 — Question 19. Find the value(s) of k such that

a) $(-15, 9)$, $(3, 3)$ and $(k, -5)$ are collinear.

b) $(1, k)$, $(-3k, 3)$ and $(7, \frac{3}{2})$ are collinear.

1.03 — Question 20. Three points A, B and C have coordinates $(-3, 10)$, $(5, 2)$ and $(9, 6)$ respectively.



a) Find the equation of the perpendicular bisector of A and B in the form $y = mx + c$.

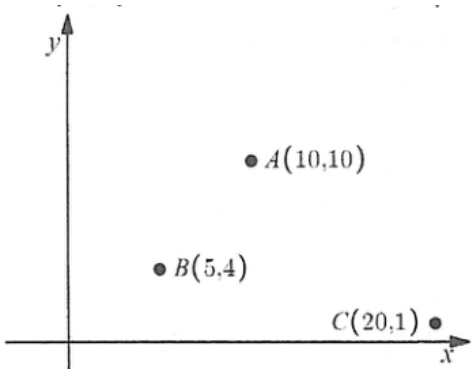
b) Find the equation of the perpendicular bisector of B and C in the form $y = mx + c$.

c) Find the point P, where the two perpendicular bisectors intersect.

d) Find the distance between the point P and each of the three points.

1.03 — Question 21. By considering the line that is perpendicular to $3y - 4x = 5$ which passes through the point $P(11, 8)$, find the shortest distance between $P(11, 8)$ and the line $3y - 4x = 5$.

1.03 — Question 22. Three points $A(10, 10)$, $B(5, 4)$ and $C(20, 1)$ and $D(15, k)$ form the vertices of a trapezium.



a) Show that AB is not perpendicular to BC.

Find

b) the equation of the line passing through B and C in the form $y = mx + c$

c) the equation of the line passing through A and D in the form $y = mx + c$ given that it is parallel to the line passing through points B and C

d) the value of k

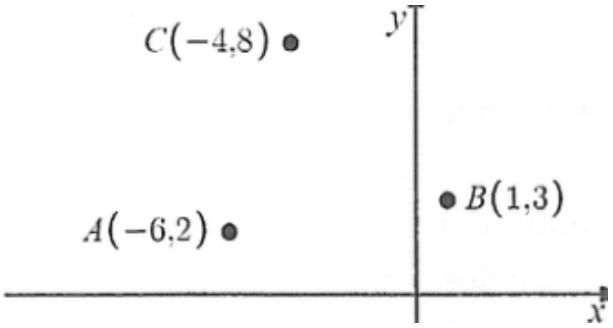
e) the distances between A & D, and B & C

f) the equation of the line which is perpendicular to BC and passes through A

g) the shortest distance from A to the line BC

h) the area of the trapezium $ABCD$

1.03 — Question 23. The points $A(-6,2)$, $B(1,3)$, $C(-4,8)$ and D form a quadrilateral.



a) Show that the distances between A and B, and between B and C are the same.
The line AC is perpendicular to the line BD.
b) Show that the possible coordinates of the point D satisfy the line $x + 3y = 10$.

1.03 — Question 24. Four points $A(-3,-1)$, $B(2,-\frac{3}{5})$, $C(4,4)$ and $D(-1,\frac{18}{5})$ form a quadrilateral. By considering the diagonals only, AC and BD, determine its type.

1.12 — Question 1. State the centre and radius of the following circles:

a) $(x-2)^2+(y-7)^2=3^2$
b) $x^2+(y+2)^2=3.5^2$
c) $(x+8)^2+(y-5)^2=16$
d) $(x+12)^2+(y+9)^2=19$
e) $(x-3)^2+(y+16)^2=28$
f) $(x-\frac{1}{2})^2+y^2=\frac{9}{4}$
g) $(x+2k)^2+(y-k+1)^2=100k,\ k>0$

1.12 — Question 2. Write down the equation of the circle in the form $(x+a)^2+(y+b)^2=c$, where $c>0$, given the following centre and radius:

a) Centre $(-3,-8)$, Radius 9
b) Centre $(4,-3)$, Radius 5
c) Centre $(0,0)$, Radius 100
d) Centre $(6,1)$, Radius $\sqrt{17}$
e) Centre $(-11,6)$, Radius $2\sqrt{2}$
f) Centre $(-\frac{2}{3},-\frac{3}{2})$, Radius $\frac{4}{9}$
g) Centre $(k-5,7-k)$, Radius $(k+2),\ k>0$

1.12 — Question 3. Write down the exact area of these circles:

a) $(x+4)^2+(y-1)^2=5^2$
b) $x^2+(y+\frac{3}{8})^2=\frac{49}{9}$
c) $3(x+7)^2+3y^2=363$

1.12 — Question 4. Write the equations for each of these circles in the form $x^2+ax+y^2+by+c=0$:

a) $x^2+(y+6)^2=2$
b) $(x-10)^2+y^2=130$
c) $(x+2)^2+(y-8)^2=66$
d) $(x-11)^2+(y-15)^2=1600$
e) $(x+4k)^2+(y-2k)^2=20k^2$

1.12 — Question 5. Write the equations for each of these circles in the form $(x+a)^2+(y+b)^2=c$, where $c>0$:

a) $x^2+2x+y^2+6y-8=0$
b) $x^2-8x+y^2-20y-784=0$
c) $x^2+y^2+6x-14y-82=0$
d) $x^2+y^2-5x-9y-\frac{57}{2}=0$
e) $x^2-\frac{2}{3}x+y^2+\frac{4}{9}y-\frac{230}{81}=0$
f) $x^2+y^2-\frac{3}{2}x-\frac{5}{4}y-\frac{1091}{64}=0$
g) $x^2+y^2+\frac{14}{5}x+\frac{5}{2}y+\frac{1009}{400}=0$
h) $x^2+y^2+kx-6ky+3k=0,\ k>\frac{12}{37}$

1.12 — Question 6. Determine whether the following equations define a circle:

a) $x^2-6x+y^2+16y+83=0$
b) $x^2+10x+y^2-y+\frac{101}{4}=0$
c) $x^2-4x-y^2-10y-46=0$
d) $x^2+12x+3y^2-6y+3=0$
e) $x^2+\frac{2}{3}x+y^2-\frac{4}{3}y-\frac{1}{3}=0$

1.12 — Question 7. Determine the possible values of k such that the equation $x^2+y^2+8x-12y-k=0$ represents a circle.

1.12 — Question 8. Determine the possible values of k such that the equation $x^2+y^2-3x+9y+\frac{k}{6}=0$ represents a circle.

1.12 — Question 9. Determine whether these circles intersect with either the x-axis or the y-axis:

a) $(x-4)^2+(y-5)^2=9$
b) $(x+1)^2+(y-6)^2=25$
c) $(x+3)^2+(y-4)^2=8$
d) $(x+8)^2+(y+3)^2=10$

1.12 — Question 10. By considering the centre and radius, determine the values of k, where $k>0$, such that these circles intersect the x-axis at two distinct points:

a) $(x-2)^2+(y-k)^2=36$
b) $(x+5)^2+(y-k)^2=10$
c) $(x-1)^2+(y-4)^2=k^2$
d) $(x+7)^2+(y+8)^2=(k+1)^2$

1.12 — Question 11. By considering the centre and radius, determine the values of k, where $k > 0$, such that these circles intersect the y-axis at two distinct points:

a) $(x-k)^2 + y^2 = 49$
b) $(x+k)^2 + (y-4)^2 = 9$
c) $(x-6)^2 + (y+3)^2 = k^2$

1.12 — Question 12. Sketch the following circles, identifying their centre and the coordinates of the point on the circle that is closest to the x-axis:

a) $(x-3)^2 + (y+5)^2 = 4$
b) $(x-6)^2 + (y-3)^2 = 4$
c) $(x+7)^2 + (y+5)^2 = 16$
d) $(x+6)^2 + (y-6)^2 = 10$

1.12 — Question 13. Sketch the following circles, identifying their centre and where they intersect the coordinate axes:

a) $x^2 + (y-3)^2 = 9$
b) $x^2 + (y+2)^2 = 16$
c) $(x-5)^2 + y^2 = 81$
d) $(x+5)^2 + y^2 = 25$

1.12 — Question 14. Sketch the following circles, identifying their centre and where they cross the coordinate axes:

a) $(x-4)^2 + (y-4)^2 = 16$
b) $(x+2)^2 + (y+2)^2 = 13$
c) $(x-1)^2 + (y+3)^2 = 17$
d) $(x+7)^2 + (y-2)^2 = 20$

1.12 — Question 15. Sketch the following circles, identifying their centre and the exact points at which they cross the coordinate axes:

a) $(x+3)^2+(y-8)^2=16$
b) $(x+7)^2+(y+5)^2=36$
c) $(x-8)^2+(y-2)^2=81$
d) $(x-1)^2+(y-\frac{11}{16})^2=\frac{697}{256}$

1.12 — Question 16. Determine whether the point $(3,7)$ is inside, on, or outside of these circles:

a) $(x-1)^2+(y-5)^2=9$
b) $(x-3)^2+(y-9)^2=4$
c) $(x-5)^2+(y-6)^2=1$

1.12 — Question 17. A circle has centre $(2, -2)$ and the point $(14, 14)$ lies on the circle. Find the equation of the circle.

1.12 — Question 18. The points A and B lie on a circle. The line AB passes through the centre of the circle. Find the equation of the circle in each case:

a) $A = (-5, -7)$ and $B = (9, -5)$
b) $A = (-8, 7)$ and $B = (6, -1)$
c) $A = (4, 7)$ and $B = (-4, -9)$
d) $A = (-2, -1)$ and $B = (12, 13)$

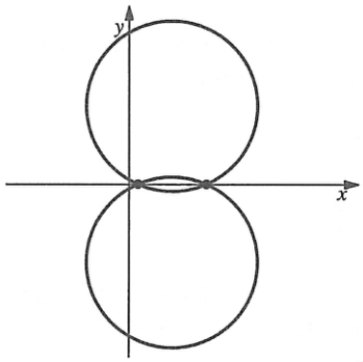
1.12 — Question 19. A circle passes through the points $A(6, 2)$ and $B(8, 4)$, and the centre of the circle lies on the x-axis. Find the equation of the circle.

1.12 — Question 20. A circle passes through the points $A(-5, 1)$ and $B(3, 11)$, and the centre of the circle lies on the y-axis. Find the equation of the circle.

1.12 — Question 21. A circle passes through the points $A(1, -6)$ and $B(17, -10)$, and the centre of the circle lies on the line $y = x$. Find the equation of the circle.

1.12 — Question 22. Show that the shortest distance from $A(-3, 8)$ to the circumference of the circle with equation $(x - 8)^2 + (y + 3)^2 = 9$ is $a\sqrt{2} + b$, where a and b are integers.

1.12 — Question 23. The diagram below shows two circles of radius 10 that pass through the points $(1, 0)$ and $(9, 0)$. Find the equations of the two circles.



1.12 — Question 24. A circle passes through the points $A(7, 2)$, $B(3, 2)$ and $C(3, -6)$. Find the equation of the circle.

1.12 — Question 25. A circle passes through the points $A(-1,-12)$, $B(0,-13)$ and $C(3,-4)$. Find the equation of the circle.

1.12 — Question 26. A circle passes through the points $A(-8,1)$, $B(-9,2)$ and $C(0,11)$. Find the equation of the circle.

1.12 — Question 27. A circle passes through the points $A(1,1)$ and $B(2,6)$. A diameter of the circle lies on the y-axis. Find the equation of the circle.

1.12 — Question 28. The points $A(-6,-1)$, $B(2,7)$ and $C(4,5)$ lie on a circle. It is given that AC is a diameter of the circle. Find the area of the triangle ABC.

1.12 — Question 29. A circle passes through the points $A(-12, 11)$, $B(-8, -1)$ and $C(-2, 1)$

a) Calculate the lengths of the sides of the triangle ABC.

b) Hence show that AC is a diameter of the circle.

c) Find the area of triangle ABC.

1.12 — Question 30. The points $A(2, 6)$ and $B(6, k)$ lie on a circle such that AB is a diameter of the circle. The line through A and B also passes through $(10, 20)$. Find the equation of the circle.

1.12 — Question 31. Determine whether the following lines intersect the circle with equation $(x+4)^2+(y-6)^2 = 20$ once, twice or not at all:

a) $y = 5x + 1$
b) $y = 1 - 4x$
c) $y = 2x + 24$

1.12 — Question 32. The line with equation $y = k$ is a tangent to the circle $x^2 + y^2 - 3x + 10y + \frac{427}{16} = 0$. Find the possible values of k.

1.12 — Question 33. The vertices of an equilateral triangle lie at $(0,0)$, $(-10,10\sqrt{3})$ and $(10,10\sqrt{3})$. A circle is drawn within the triangle so that the three sides are all tangents to the circle. Find the area of the circle.

1.12 — Question 34. The circle with equation $(x-3)^2+(y+5)^2=34$ passes through the origin. Find the gradient of the tangent to the circle at the origin.

1.12 — Question 35. The point $A(1,2)$ lies on the circle with equation $(x+2)^2+(y-3)^2=10$. Find the equation of the tangent to the circle at A in the form $y=ax+b$, where a and b are integers.

1.12 — Question 36. The point A has coordinates $(\frac{5}{2},-\frac{1}{2})$ and lies on the circle with equation $(x-\frac{1}{2})^2+(y+\frac{3}{2})^2=5$. Find the equation of the tangent to the circle at A in the form $y=ax+b$.

1.12 — Question 37. The point A has coordinates $(1, 1)$ and lies outside of the circle $(x + 3)^2 + (y - 9)^2 = 10$. The point B lies on the circle such that AB is a tangent to the circle. Find the length of AB.

1.12 — Question 38. Find the two equations of the tangents to the circle $(x - 3)^2 + (y - 4)^2 = 100$ when $x = -5$ in the form $y = ax + b$, where a and b are rational numbers.

1.12 — Question 39. Find the two equations of the tangents to the circle $(x+5)^2+(y-9)^2=10$ when $y=10$ in the form $y=ax+b$, where a and b are integers.

1.12 — Question 40. The line $x+7y=82$ is a tangent to a circle with centre $(-\frac{5}{2},\frac{17}{2})$. Find the equation of the circle.

1.12 — Question 41. A circle has equation $(x-11)^2+(y+4)^2=16$. The straight line with equation $y=kx$, where k is a constant, meets the circle.

a) Show that the coordinates of any points of intersection between the line and the circle satisfies the equation $(1+k^2)x^2+2(4k-11)x+121=0$

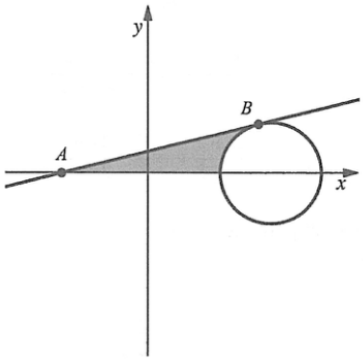
b) Hence, find the two values of k for which the line $y=kx$ is a tangent to the circle.

1.12 — Question 42. A circle has equation $(x+3)^2+(y-3)^2=80$. The straight line with equation $y=2x+k$, where k is a constant, meets the circle.

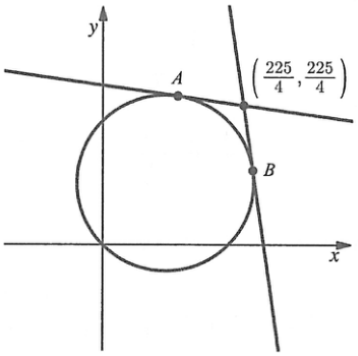
a) Show that the coordinates of any points of intersection between the line and the circle satisfies the equation $5x^2+2(2k-3)x+k^2-6k-62=0$

b) Hence, find the two values of k for which the line $y=2x+k$ is a tangent to the circle.

1.12 — Question 43. The diagram below shows the circle with equation $(x-10)^2+y^2=17$. The line passing through A and B is a tangent to the circle at the point where $x=9$. Find the area of the shaded region.



1.12 — Question 44. The diagram below shows the circle with equation $(x-25)^2+(y-25)^2=1250$. The two lines are both tangents to the circle. The tangent at A has equation $y=-\frac{1}{7}x+a$ and the tangent at B has equation $y=-7x+b$. Show that the two lines intersect at $(\frac{225}{4},\frac{225}{4})$.



1.12 — Question 45. Find the exact shortest distance from the line $y = 2x + 1$ to the circle $x^2 + (y - 7)^2 = 3$.

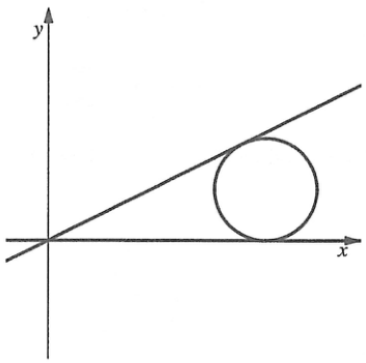
1.12 — Question 46. A circle has equation $x^2 - 14x + y^2 + 18y - 40 = 0$. The points A and B lie on the circle such that $(2, 1)$ is the midpoint of A and B. Find the shortest distance around the circumference of the circle from A to B.

1.12 — Question 47. A circle, P, has equation $x^2 + y^2 + 6y = 7$. A second circle, Q, has its centre on the x-axis and has the same radius as P. Q touches P at one point. Find the possible equations of Q.

1.12 — Question 48. A circle centred at $(8, k)$ has $3x - 2y = 12$ and $3x - 2y = 24$ as tangents. Find the equation of the circle.

1.12 — Question 49. A circle with radius k has both the x-axis and the y-axis as tangent lines. $2x+y=50$ is also a tangent line to the circle. Find the two possible exact values of k and sketch both situations on the same graph.

1.12 — Question 50. A circle of area 25π square units is drawn that has both the x-axis and $y=\frac{1}{2}x$ as tangents, as shown in the diagram below. Find the equation of the circle.



1.12 — Question 51. For the circle shown in the diagram below, determine the percentage of the circle's area that is below the x-axis.

